# Continuous attractors

## and navigation

Computational Neuroscience

University of Bristol

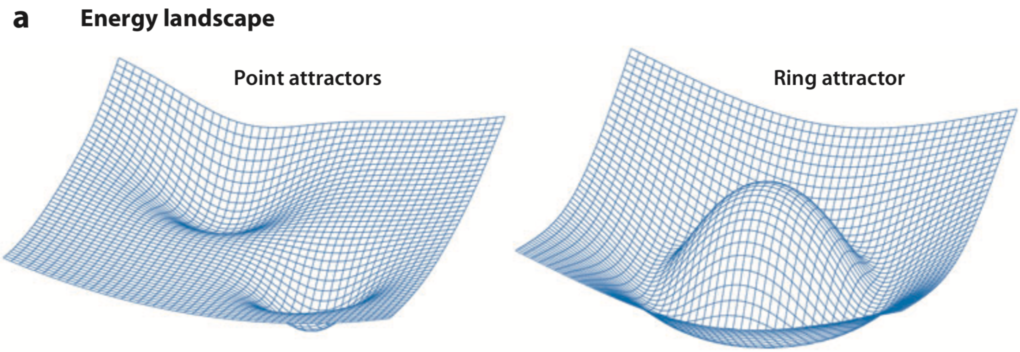
Slides adapted from L Aitchison

mrule

**Learning outcomes:**

- Head-direction cell attractors
- Grid-cell attractors

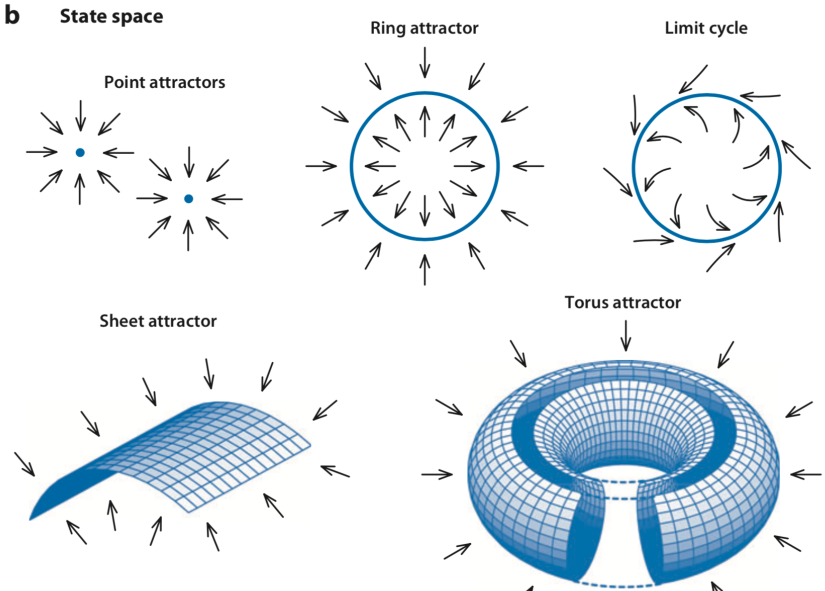
# *Attractor networks*



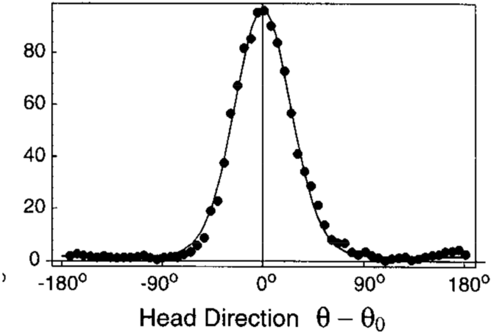

*Knierim, J.J., Zhang, K., 2012. Annu. Rev. Neurosci. 35, 267–285.*

Attractor networks are recurrent neural networks where the internal dynamics evolves towards a stable pattern.

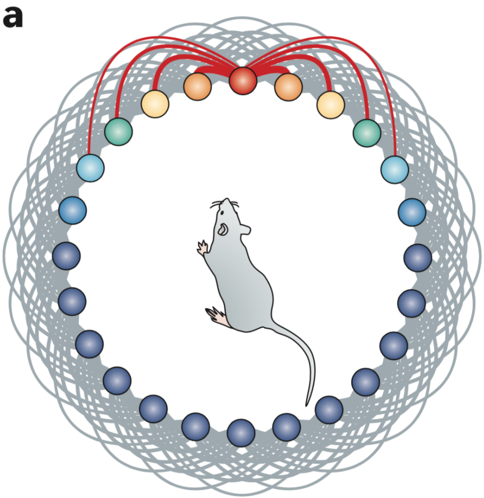
## *Attractor networks*



# Continuous attractors in subiculum




*McNaughton, et al. (2006). Nat Rev Neurosci 7, 663–678*

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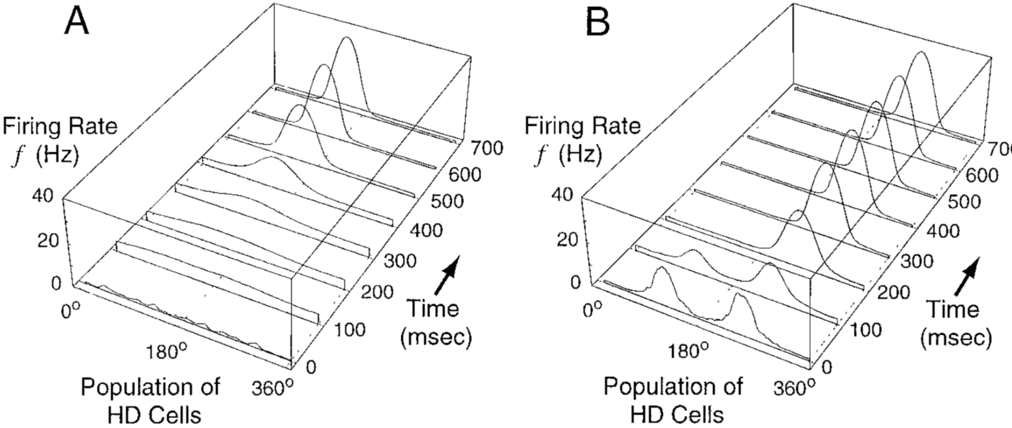
# *Continuous attractors in subiculum*

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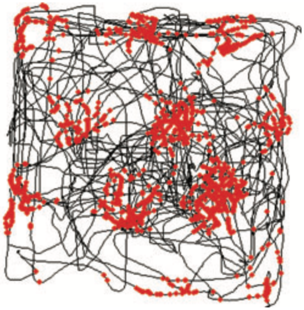
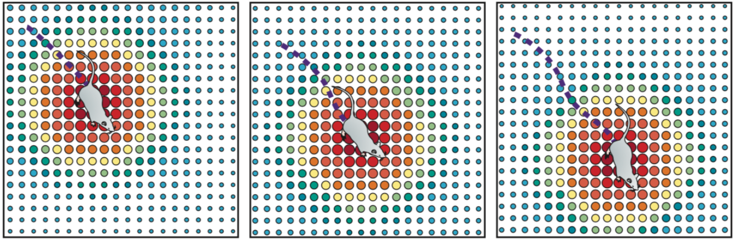
# Continuous attractors in entorhinal cortex

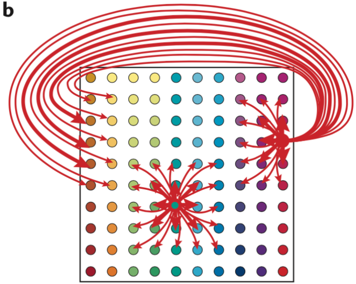
**Grid cell**

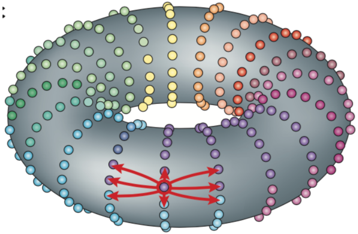


*McNaughton, et al. (2006). Nat Rev Neurosci 7, 663–678*

# *Continuous attractors in entorhinal cortex*

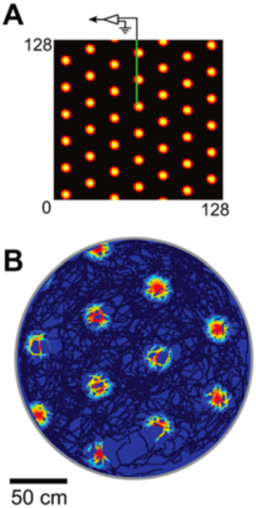




McNaughton, et al. (2006). Nat Rev Neurosci 7, 663–678

# Continuous attractors in entorhinal cortex




Network activity

Single cell activity as a function of location

*Burak and Fiete, PLoS Comput Biol (2009)*

# *Ring attractor math*

Let

- $\psi \in [0,, 2\pi)$ be a circular (ring) variable,
- $\boldsymbol{x}(\psi)$ be the activity of a population of neurons with preferred reponses around this ring.

We will explore the rate model

$$\dot{\boldsymbol{x}}(\psi, t) = -\boldsymbol{x}(\psi, t) + \sigma\{[\boldsymbol{k} * \boldsymbol{x}](\psi, t) - \vartheta\}$$

Where

$$\boldsymbol{k}(\psi) = \cos(\psi)$$

Study steady-state solutions $\dot{\boldsymbol{x}}(\psi, t) = 0$

$$\boldsymbol{x}(\psi, t) = \sigma\{[\boldsymbol{k} * \boldsymbol{x}](\psi, t) - \vartheta\}$$

Is a centre-surround synaptic coupling kernel

# References

Burak, Y., Fiete, I.R., 2009. Accurate path integration in continuous attractor network models of grid cells. PLoS Comput Biol 5, e1000291.

McNaughton, B.L., Battaglia, F.P., Jensen, O., Moser, E.I., Moser, M.-B., 2006. Path integration and the neural basis of the 'cognitive map'. Nat Rev Neurosci 7, 663-678.

O'Reilly, R.C., McClelland, J.L., 1994. Hippocampal conjunctive encoding, storage, and recall: avoiding a trade-off. 4, 661–682.

Zhang, K., 1996. Representation of spatial orientation by the intrinsic dynamics of the head-direction cell ensemble: a theory. Journal of Neuroscience 16, 2112–2126.

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